Hospital Emergency Room Analysis Dashboard

An interactive Power BI dashboard designed to understand patient flow, operational performance, waiting time, satisfaction, and patient demographics in an Emergency Room environment.

1. Project Overview

Emergency Room data can contain thousands of patient records, but raw records alone do not provide a clear view of operational performance.

This project transforms patient-level data into an interactive Power BI dashboard that brings key metrics, trends, and patient patterns into one place.

The analysis focuses on:

- Patient visit volume
- Average waiting time
- Patient satisfaction
- Patient demographics
- Department-level waiting time
- Daily and hourly patient flow
- Month-over-month performance

The goal was to create a dashboard that turns raw patient data into clear operational insights and makes important patterns easier to identify and investigate.

2. Business Questions

The dashboard was built around a few practical questions:

- How many patients are visiting the ER?
- How is patient volume changing over time?
- What is the average waiting time?
- How does current performance compare with the previous month?
- Which departments have higher average waiting times?
- When is patient traffic at its highest?
- How do visits and satisfaction vary across demographic groups?

3. Dataset

The dataset contains **9,216 patient records** across **11 original fields**.

Field	Description
Date	Patient visit date and time
Patient ID	Unique patient identifier
Gender	Patient gender
Age	Patient age
Satisfaction Score	Recorded patient satisfaction score
First Name	Patient first name
Last Name	Patient last name
Race	Patient race/ethnicity category
Patient Admin Flag	Indicates whether the patient was admitted
Wait Time	Patient waiting time
Department Referral	Department associated with the visit

The dataset combines operational metrics, such as wait time and department referral, with patient attributes such as age, gender, race, and satisfaction score.

4. Data Preparation & Cleaning

Before building the dashboard, the raw data was prepared in **Power Query.** The purpose of this stage was to make the dataset consistent, analysis-ready, and suitable for time-based reporting.

Date & Time Transformation

The original Date field contained date and time information.

The following transformations were performed:

- Converted the Date field into an appropriate Date/Time format.
- Created a separate **Date** field.
- Created a separate **Time** field.
- Created **Start of Hour** for hourly patient-flow analysis.

These transformations made it possible to analyze patient traffic at different levels of time granularity.

Data Type Standardization

Several fields were standardized before analysis-

- Converting Age from text to whole number.
- Created Age Groups to support demographic analysis.
- Standardizing Gender values.
- Creating a Full Name field by combining First Name and Last Name.

Date Table & Time Intelligence

A dedicated **Date Table** was created to support time-based analysis and month-over-month comparisons. This was important because the dashboard uses calculations such as:

- Previous Month
- Month-over-Month Variance
- Growth %
- Day of Week
- Daily trends

A Day of Week column was also created using DAX to maintain the correct weekday order in visualizations.

5. Data Modeling & DAX

The project uses a structured Power BI data model with:

- **Hospital_ER** as the main fact table.
- **Date Table** as the supporting date dimension.

The Date Table provides the calendar context required for daily and monthly analysis.

This structure allows DAX measures to respond dynamically to filters and time selections rather than relying on static calculations.

Core KPIs

```
Patients Visit =
COUNTROWS(Hospital_ER)

Avg Wait Time =
AVERAGE(Hospital_ER[Wait Time])

Avg Satisfaction Score =
AVERAGE(Hospital_ER[Satisfaction Score])
```

Additional measures were created for:

- Previous Month
- Variance
- Growth %
- Maximum/Minimum values
- Busiest Day
- Busiest Time

The dashboard also uses dynamic indicators to show whether key metrics have increased or decreased compared with the previous month.

6. Dashboard Analysis

Patient Volume & Trends

Daily trend analysis helps identify changes in patient volume over time. Maximum and minimum points are dynamically highlighted to make important changes easier to spot.

Waiting Time

Average waiting time is monitored overall and compared across departments to identify differences in operational performance.

Patient Satisfaction

Average satisfaction is tracked over time and can be explored across different demographic dimensions.

Demographic Analysis

Patients are analyzed by:

- Gender
- Age Group
- Race

A **Field Parameter** allows the user to dynamically switch between these dimensions within the same visual, keeping the dashboard interactive without adding unnecessary charts.

Patient Flow

Patient traffic is analyzed by both **day of week and hour of day**. A day-hour matrix provides an additional view of when patient demand is concentrated.

7. Key Findings

Patient Volume

There are **9,216 patient records** in the dataset.

The gender distribution is relatively balanced:

- Male: **4,705**
- Female: **4,487**
- Not Confirmed: **24**

This suggests that the overall patient volume is not heavily skewed toward one gender.

Age Distribution

The largest age segment is **Adults aged 18–64**, with approximately **5,509 records**.

This is followed by:

- Children: **1,971**
- Seniors 65–79: **1,736**

The adult population therefore represents the largest share of the dataset.

Waiting Time

The overall average waiting time is approximately **35.3 minutes**.

Department-level averages are relatively close, ranging roughly from **34.7 to 36.8 minutes** among the named departments.

This suggests that waiting-time differences between departments are present, but they are not dramatically separated in this dataset.

Satisfaction

The average recorded satisfaction score is approximately **5.0**.

However, an important data-quality consideration is that many records do not contain a satisfaction score.

Therefore, the average satisfaction KPI represents **records with a recorded satisfaction score**, rather than every patient visit.

This distinction is important when interpreting the KPI.

Patient Flow

The busiest weekday in the full dataset is **Monday**, with **1,377 visits**.

Saturday follows with **1,332 visits.**

The highest-volume hour in the underlying dataset is approximately **11 PM**, with **436 visits**.

These patterns demonstrate why day-and-hour analysis is useful for understanding patient flow.

8. Patient Details

Alongside the analytical dashboard, a dedicated **Patient Details** page allows users to move from aggregated insights to individual patient records.

This creates a simple analytical flow:

Summary → Pattern → Detail

Users can first identify an issue or pattern and then explore the underlying records when needed.

9. Tools & Skills

Tools:

Power BI · Power Query · DAX

Key Skills Demonstrated:

- Data Cleaning & Transformation
- Data Modeling
- DAX & Time Intelligence
- KPI Development
- Trend Analysis
- Demographic Analysis
- Interactive Dashboard Design
- Dynamic Parameters
- Data Storytelling

10. Business Impact & Recommendations

The dashboard turns raw Emergency Room data into a structured view of **patient demand, waiting time, satisfaction, demographics, and operational patterns**. This makes it easier to identify areas that may require closer monitoring and further investigation.

Key Recommendations

Review High-Demand Periods

Monday records the highest overall patient volume, while 11 PM is the busiest hour in the dataset. These patterns suggest that **staffing and resource allocation during high-demand periods could be reviewed** to better align capacity with patient flow.

Monitor Waiting Time Across Departments

Department-level average waiting times are relatively close, but continuous monitoring can help identify whether specific departments develop higher waiting times over time.

Track Patient Satisfaction Alongside Operational Metrics

Satisfaction should be monitored together with waiting time and patient volume to understand whether changes in operational performance are associated with changes in patient experience.

Use Demographic Patterns for Deeper Analysis

Adults aged 18–64 represent the largest patient group. Further analysis of demographic patterns can help understand whether different patient groups show different visit or satisfaction patterns.

Business Impact

Overall, the dashboard provides a **data-driven starting point for operational monitoring and planning**. It helps decision-makers move from simply viewing patient records to identifying demand patterns, monitoring performance, and determining where further investigation may be valuable.